



DEPARTMENT OF COMPUTER STUDIES

SURVEY FINDINGS SUMMARY

Flood Susceptibility and Awareness Survey

Total Respondents: 50 | Data Gathering Period: May 05–12, 2026
Sampling: Purposive-Convenience | 12 Barangays across Bacoor City

1. Flood Experience and Hazard Exposure

Key Finding	Detail
92% of respondents experience flooding	Occasionally + Frequently + Very Frequently combined
54% experience flooding almost every rainy season	"Very Frequently" — highest frequency category
82% report moderate to severe flooding	Knee-deep or higher (Moderate 24% + Severe 58%)
58% report severe flooding	Waist-deep or higher — highest severity category

Q1. Barangay of Residence

Barangay	No. of Respondents
Zapote 1	15
Mambog 1	12
Niog 3	5
Niog 2	3
Niog 1	2
San Nicolas 3	3
Molino 1	2
Talaba 3	2
Talaba 1	2
Talaba 2	1
Habay	1
Queens Row West	1
Detachment	1
TOTAL	50

Top flood effects reported by respondents (Q8 — select all that apply):

Effect	Count	% of Respondents
Disruption to daily activities (work, school)	48	96.0%
Health risks	42	84.0%
Property damage	39	78.0%
Displacement or evacuation	19	38.0%
Loss of livelihood	18	36.0%

2. Flood Risk Awareness and Warning Channels**Flood area awareness (Q7):**

Awareness Level	Count	%
Well-informed	10	20.0%
Somewhat aware	30	60.0%
Not sure	4	8.0%
Not aware at all	6	12.0%

Key Finding: Most respondents (80%) reported being only somewhat aware, unsure, or not aware at all regarding flood-prone areas, while only 20% considered themselves well-informed. This suggests that many respondents may have limited knowledge about flood susceptibility within their area.

Primary warning channels (Q6 — select all that apply):

Channel	Count	% of Respondents
Social media	33	66.0%
Text/SMS alerts	32	64.0%
Neighbors / word of mouth	22	44.0%
News/TV	21	42.0%
Barangay PA system	15	30.0%
I receive no warning	2	4.0%

Key Finding: Official digital channels such as social media posts from government agencies (66%) and SMS alerts (64%) are the primary warning sources among respondents. Meanwhile, only 30% receive alerts through the Barangay PA system, suggesting stronger dependence on digital dissemination methods than traditional on-site warning mechanisms.

3. Flood Information Quality (Likert Scale, Q9–Q11)

Scale: 1 = Very Dissatisfied / Never | 3 = Neutral / Sometimes | 5 = Very Satisfied / Always

Question	Mean Score	Interpretation
Q9. Satisfaction with flood information	2.62	Neutral / Sometimes
Q10. Timeliness of flood warnings	2.70	Neutral / Sometimes

Question	Mean Score	Interpretation
Q11. Clarity of flood risk communication	2.72	Neutral / Sometimes

Key Finding: All three Likert items yielded mean scores of 2.62–2.72, indicating that respondents are generally dissatisfied with the current quality, timeliness, and clarity of flood information.

4. Decision-Making Due to Lack of Flood Information (Q12–13)

Top Problems in Getting Flood Info (Q12)	Frequency	% (n=50)
Late / Delayed Information	13	26%
Lack of Information / No Warnings	10	20%
Unclear or Inaccurate Information	6	12%
Sudden / Unpredictable Flooding	4	8%
Lack of Location-Specific Risk Info	3	6%

Wrong Decisions Due to Lack of Flood Info (Q13)	Count	%
YES — made a wrong decision	32	64%
No	18	36%

Key Finding: 64% of respondents (32 of 50) have made a wrong decision due to insufficient flood information — providing strong empirical support for the need for a reliable, accessible flood risk information tool.

5. Proposed Solution Concept (Working Title)

Working Title: FloodSense: An Augmented Reality Mobile Application with Machine Learning-Based Flood Susceptibility Mapping for Bacoar City

Based on the identified gaps, this study proposes developing a mobile application that provides residents with location-specific flood susceptibility information for Bacoar City. Using a machine learning model trained on rainfall data, elevation, soil type, drainage proximity, and historical flood records, the app classifies flood risk into four clear levels: Low, Moderate, High, and Very High - for each zone/ barangay in the city.

Residents open the app, set their own parameters — how much rain is expected and how long it will fall — then point their phone camera at any flat surface. The app renders a color-coded flood susceptibility map anchored to the ground through augmented reality, giving them an intuitive, visual answer to the question: "Is my area at risk right now?"